

LOW ACRYLONITRILE NBR: SOLUTION-POLYMERIZED NITRILE RUBBER WITH CONTROLLED ACRYLONITRILE CONTENT

1. ABSTRACT

This report reviews patent evidence relevant to low-acrylonitrile nitrile butadiene rubber (NBR), with emphasis on synthesis routes that control acrylonitrile incorporation during copolymerization of butadiene and acrylonitrile. The primary patent, CN115725020A, discloses a solution polymerization process designed to avoid soap-derived impurities typical of emulsion NBR and to produce a colorless, transparent raw rubber with improved processing and vulcanizate properties. The patent positions solution polymerization as an industrially viable alternative to emulsion routes, using a mixed solvent system composed of a benzene solvent and a ketone solvent, with cooling before initiation and subsequent radical polymerization under low temperature and moderate pressure.

The disclosed composition window for the monomer feed is broad, with butadiene and acrylonitrile mass percentages of 60-80% and 40-20%, respectively, which encompasses low-acrylonitrile grades at the lower end of acrylonitrile content. The examples show that varying the acrylonitrile charge and solvent/co-solvent selection changes the combined acrylonitrile content in the raw rubber from 27.3% to 41.1%, while maintaining high monomer conversion (>85% in the solution examples). The process also uses chain-transfer control via tert-dodecyl mercaptan and initiators such as PMHP or PHP, with reaction temperatures below 8°C and reaction times of 12-14 h.

Overall, the patent landscape in the provided manifest is limited to one primary patent, but the evidence strongly indicates that low-ACN NBR can be approached through solution polymerization with careful monomer ratio control, solvent selection, low-temperature initiation, and mercaptan-mediated molecular-weight regulation. The key trend is the replacement of emulsion polymerization with a cleaner solution process to improve purity and downstream vulcanizate performance while retaining high conversion and tunable acrylonitrile incorporation.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: CN115725020A
- Patent Title: Method for preparing nitrile rubber through solution polymerization
- Assignee: Not disclosed
- Jurisdiction: CN
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of nitrile rubber by cooling a mixed solution of butadiene and acrylonitrile, then initiating polymerization with an initiator in a mixed solvent system composed of a benzene solvent and a ketone solvent. The process is designed to produce soap-free NBR with reduced organic impurities and improved processing and vulcanizate performance.

- Polymerization kettle volume: 5 L — Example 1 - Toluene: 1.6 L — Example 1 - Methyl ethyl ketone: 1.5 L — Example 1 - EDTA iron sodium salt aqueous solution: 4.1 wt% , 3.5 mL — Example 1 - Sodium carbonate aqueous solution: 10 wt% , 10 mL — Example 1 - Sodium dithionite aqueous solution: 0.4 wt% , 3.5 mL — Example 1 - Dithionite aqueous solution: 6.2 wt% , 8 mL — Example 1 - tert-Dodecyl mercaptan: 6 mL — Example 1 - Butadiene: 850 mL — Example 1 - Acrylonitrile: 200 g — Example 1 - Cooling bath temperature: 5°C — Example 1 - Material temperature after cooling: 5-8°C — Example 1 - PMHP: 0.88 mL — Example 1 - Pressure: 0.2-0.5 MPa — Example 1 - Polymerization temperature: <8°C — Example 1 - Reaction time: 12.5 h — Example 1 - Antioxidant 1520: 2.5 g — Example 1 - Steam coagulation temperature: 98°C — Example 1 - Mooney viscosity: 48.6 — Example 1 - Monomer conversion: 85.7% — Example 1 - Molecular mass distribution index: 3.7 — Example 1 - Vinyl unit content: 11.6% — Example 1 - Combined acrylonitrile content: 27.3% — Example 1 - Toluene: 1.8 L — Example 2 - Methyl ethyl ketone: 1.3 L — Example 2 - Dodecyl mercaptan: 5.8 mL — Example 2 - PMHP: 0.90 mL — Example 2 - Acrylonitrile: 230 g — Example 2 - Butadiene: 840 mL — Example 2 - Reaction time: 13 h — Example 2 - Monomer conversion: 86.4% — Example 2 - Molecular mass distribution index: 4.2 — Example 2 - Mooney viscosity: 53.8 — Example 2 - Vinyl unit content: 12.6% — Example 2 - Combined acrylonitrile content: 30.2% — Example 2 - Ethylbenzene: 2.0 L — Example 3 - Acetone: 1.2 L — Example 3 - Dodecyl mercaptan: 5.7 mL — Example 3 - PHP: 1.0 mL — Example 3 - Acrylonitrile: 260 g — Example 3 - Butadiene: 830 mL — Example 3 - Reaction time: 12 h — Example 3 - Monomer conversion: 87.4% — Example 3 - Molecular mass distribution index: 3.7 — Example 3 - Mooney viscosity: 58.4 — Example 3 - Vinyl unit content: 11.5% — Example 3 - Combined acrylonitrile content: 33.6% — Example 3 - Benzene: 1.7 L — Example 4 - Acetone: 1.4 L — Example 4 - Dodecyl mercaptan: 5.5 mL — Example 4 - PHP: 1.1 mL — Example 4 - Acrylonitrile: 280 g — Example 4 - Butadiene: 800 mL — Example 4 - Reaction time: 12 h — Example 4 - Monomer conversion: 87.7% — Example 4 - Molecular mass distribution index: 4.3 — Example 4 - Mooney viscosity: 63.6 — Example 4 - Vinyl unit content: 11.2% — Example 4 - Combined acrylonitrile content: 35.4% — Example 4 - Chlorobenzene: 1.9 L — Example 5 - Methyl ethyl ketone: 1.2 L — Example

5 - Dodecyl mercaptan: 5.4 mL — Example 5 - PMHP: 1.2 mL — Example 5 - Acrylonitrile: 325 g — Example 5 - Butadiene: 750 mL — Example 5 - Reaction time: 14 h — Example 5 - Monomer conversion: 88.7% — Example 5 - Molecular mass distribution index: 4.4 — Example 5 - Mooney viscosity: 72.9 — Example 5 - Vinyl unit content: 10.7% — Example 5 - Combined acrylonitrile content: 41.1% — Example 5 - Butadiene and acrylonitrile mass percentage: 60-80% / 40-20% — Claims - Mixed solvent to monomer mass ratio: 3.0-4.0 kg/kg — Claims - Benzene solvent types: benzene, toluene, xylene, ethylbenzene, chlorobenzene — Claims - Ketone solvent carbon number: 3-5 carbon atoms — Claims

Relevant Experimental Evidence - Example 1: Baseline solution polymerization in toluene/methyl ethyl ketone at <8°C gave 85.7% conversion, Mooney viscosity 48.6, and combined acrylonitrile content 27.3%. - Example 2: Increasing acrylonitrile charge to 230 g and adjusting solvent/mercaptan/initiator levels maintained high conversion (86.4%) and raised combined acrylonitrile content to 30.2%. - Example 3: Ethylbenzene/acetone solvent system with 260 g acrylonitrile achieved 87.4% conversion and 33.6% combined acrylonitrile content. - Example 4: Benzene/acetone system with 280 g acrylonitrile produced 87.7% conversion and 35.4% combined acrylonitrile content. - Example 5: Chlorobenzene/methyl ethyl ketone system with 325 g acrylonitrile gave the highest conversion in the examples (88.7%) and the highest combined acrylonitrile content (41.1%).

Technical Relevance This patent is directly relevant because it provides a solution-polymerized NBR route that controls acrylonitrile incorporation through monomer feed and solvent selection, while avoiding emulsion-derived soap impurities. Although the examples span moderate to relatively high acrylonitrile contents, the disclosed composition window and process controls are applicable to low-acrylonitrile NBR design.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The evidence supports solution polymerization as a practical route to cleaner NBR with tunable acrylonitrile content and improved processing properties. For low-acrylonitrile NBR, the most relevant controls are the butadiene/acrylonitrile feed ratio, mixed benzene/ketone solvent selection, low-temperature initiation below 8°C, and mercaptan-based molecular-weight regulation. The patent demonstrates that high monomer conversion (>85%) is achievable while varying acrylonitrile charge and solvent system, making it a useful process basis for low-ACN NBR synthesis and scale-up.

5. REFERENCES

- CN115725020A | Method for preparing nitrile rubber through solution polymerization | Not disclosed | CN | 2023 | <https://patents.google.com/patent/CN115725020A/en>